

SAR Imaging Based on OFDM Waveform for UAV-ISAC Systems

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Abstract—Synthetic aperture radar (SAR) imaging can be effectively incorporated into unmanned aerial vehicle (UAV)-enabled integrated sensing and communication (ISAC) systems, exhibiting considerable potential for advancement in the low-altitude economy. To improve the imaging quality, we propose a probabilistic constellation shaping (PCS) method for orthogonal frequency division multiplexing (OFDM)-based UAV-ISAC imaging systems, achieved by regulating the amplitude randomness of communication symbols. Specifically, we employ the expectation of the integrated sidelobe level correlated with the periodic auto-correlation function as the imaging metric to enhance quality under symbol probability and communication achievable information rate constraints. Subsequently, the alternating optimization framework is presented to tackle the nonconvex problem. Finally, numerical simulations demonstrate the effectiveness of the proposed PCS method in enhancing SAR imaging performance.

Index Terms—Integrated sensing and communication system, orthogonal frequency division multiplexing waveform, synthetic aperture radar imaging.

I. INTRODUCTION

Unmanned aerial vehicle (UAV)-enabled integrated sensing and communication (ISAC) systems have attracted considerable interest in next-generation wireless networks [1], [2]. Serving as a promising technology for the low-altitude economy, such systems are capable of enhancing the utilization efficiency of hardware and wireless resources. Moreover, integrated with synthetic aperture radar (SAR), the UAV-ISAC system can flexibly acquire images under all-time and low-light conditions, making it suitable for various remote sensing missions [3]. In order to meet the requirement of UAV-ISAC systems for simultaneously implementing dual functionalities on a single device with high resource utilization efficiency, waveform design is a direct and effective technical approach.

Orthogonal frequency division multiplexing (OFDM) is an optimal candidate waveform for UAV-ISAC systems, widely adopted in communication systems for its high spectral efficiency [4]. Furthermore, its application for SAR imaging has also been widely investigated. In [5], a pioneering algorithm that employs OFDM signals equipped with sufficient cyclic

prefix (CP) for SAR imaging applications is presented. Nevertheless, the inherent randomness associated with the transmission of communication symbols makes it challenging to obtain high-quality SAR images via OFDM communication signals, necessitating performance trade-offs between SAR imaging and communication in OFDM signal design. For instance, authors in [4] proposed an optimal power allocation strategy for the joint communication and SAR (JCASAR) system based on OFDM waveforms, considering the randomness of data. In our previous work [6], we proposed an OFDM waveform design for JCASAR systems with subcarrier assignment and power allocation (SAPA). In this scheme, constant-envelope phase-shift keying (PSK) modulation is adopted for communication function to avoid adverse effects on imaging, yet its low data rate restricts applicable communication scenarios. In addition to the SAPA methods, research efforts have also focused on optimizing the constellation design of communication symbols to achieve a balanced ISAC performance [7], [8].

Inspired by the aforementioned observations, we propose a probabilistic constellation shaping (PCS) strategy for OFDM-based UAV-ISAC imaging systems. Specifically, we establish a multi-constrained PCS optimization model to minimize the expectation of the integrated sidelobe level (EISL) associated with the periodic auto-correlation function (P-ACF), under the constraints of symbol probability and the communication achievable information rate (AIR). To address this nonconvex problem, we present the alternating optimization (AO) algorithm, which decomposes the original problem into a series of convex subproblems that can be solved via iterative computation. Eventually, extensive numerical results verify the effectiveness of the proposed method.

II. SYSTEM MODEL AND METRICS

In this section, the monostatic UAV-ISAC geometry described in Fig. 1 is considered, where a UAV performs SAR imaging of the target area while transmitting the acquired information to communication users. Specifically, assume that the slow time is denoted by t , and the coordinate of the UAV with a velocity of v_U moving along the y -axis can be expressed as $(0, y_U(t), H_U)$, where $y_U(t) = v_U t$.

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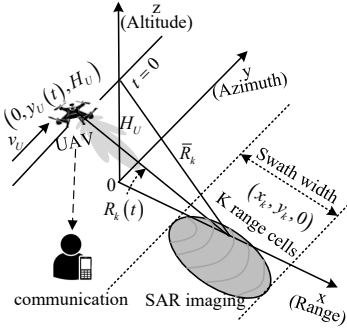


Fig. 1. The monostatic UAV-ISAC geometry.

Let the variable τ represent the fast time, and the transmitted OFDM signal with K subcarriers can be written as

$$s_t(\tau) = \sum_{k=0}^{K-1} S_{t,k} \exp(j2\pi k \Delta f \tau) \quad (1)$$

where $S_{t,k}$ denotes the k -th randomness communication symbol modulated by PSK or quadrature amplitude modulation (QAM) in slow time t . For convenience, the OFDM signal is expressed in discrete form $\mathbf{s} \in \mathbb{C}^{N \times K}$ where the slow time dimension and the fast time dimension are divided into N and K discrete time points, respectively. In order to avoid the inter-range-cell interference, we set the CP length of OFDM signals as $K - 1$ [5].

A. SAR Imaging Metric - EISL

At the SAR receiver, the received sequence after down-conversion and CP removal can be regarded as [5],

$$\mathbf{y} = \begin{bmatrix} s_0 & s_{K-1} & \cdots & s_1 \\ s_1 & s_0 & \cdots & s_2 \\ \vdots & \vdots & \ddots & \vdots \\ s_{K-1} & s_{K-2} & \cdots & s_0 \end{bmatrix} \begin{bmatrix} d_0 \\ d_1 \\ \vdots \\ d_{K-1} \end{bmatrix} + \mathbf{n}, \quad (2)$$

where $d_k = g_k \varepsilon_a(t) \exp(-j4\pi f_c R_k(\eta)/c)$ represents the weighting radar cross section coefficient, g_k and $\varepsilon_a(t)$ are the range and azimuth envelope, f_c denotes the carrier frequency and $\mathbf{n}$ is the noise.

As shown in Fig. 2, the SAR imaging processing flow mainly includes the range correlation, the range cell migration correction (RCMC) and the azimuth compression [9]. $G(f_r)$ and $H_{az}(f_t)$ are the RCMC-related multiplier and the azimuth matching filter, respectively. It is evident that the transmitted signal only significantly affects the range correlation process, which is characterized by the P-ACF in CP-OFDM signals.

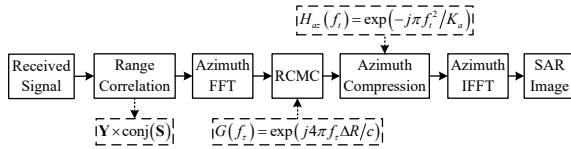


Fig. 2. The SAR imaging processing flowchart.

The P-ACF r of the CP-OFDM signal is expressed as,

$$r_k = \sum_{l=0}^{K-1} |S_l|^2 e^{j2\pi kl/K}, k = 0, \dots, K-1. \quad (3)$$

During the range correlation process, suppressing the sidelobe level of the P-ACF is critical to improving the detection of weak targets near strong targets. Given the randomness of transmitted OFDM-ISAC signals, the EISL acts as a robust metric to characterize the statistical properties of the sidelobe levels of the P-ACF, which is defined as [10],

$$\text{EISL} = \sum_{k=1}^{K-1} \mathbb{E}[|r_k|^2] = K(K-1) \left(\mathbb{E}[|S|^4] - 1 \right), \quad (4)$$

where $\kappa = \mathbb{E}[|S|^4] = \sum_{m=1}^M p_m |S_m|^4$ denotes the kurtosis of the constellation. The communication symbol S follows a constellation with M possible values and input probability $\mathbf{p}$. Evidently, a lower kurtosis yields a reduced EISL, thus improving range collection performance. Owing to the constant modulus characteristic of PSK symbols, minimal kurtosis $\kappa = 1$ can be achieved.

B. Symbol Metric - Probability Characteristics

Based on previous analysis, we have found that the input probability vector $\mathbf{p}$ of a given constellation can be optimized to reduce the kurtosis, thereby improving the imaging performance. Hence, the following constraints need to be met when designing the probabilistic constellation shape,

$$\begin{aligned} C_1 : \sum_{m=1}^M p_m &= 1, p_m \geq 0, \\ C_2 : \mathbb{E}[|S|^2] &= \sum_{m=1}^M p_m |S_m|^2 = 1, \\ C_3 : \mathbb{E}[S] &= \sum_{m=1}^M p_m S_m = 0. \end{aligned} \quad (5)$$

Notably, the complex constraint C_3 implies that the constellation points on each concentric circle have the same probability.

C. Communication Metric - AIR

The mutual information (MI) between the received communications signal $\mathbf{Y}$ and the transmitted signal $\mathbf{S}$, which consists of K parallel subchannels, directly defines the AIR, equivalent to spectral efficiency (in bps/Hz) [7],

$$I(\mathbf{S}; \mathbf{Y}) = \sum_{k=0}^{K-1} I(S_k; Y_k) = KI(S; Y), \quad (6)$$

where $I(S_k; Y_k)$ represents the MI of the k -th subchannel. Since the same constellation codebook is used for each subchannel, we only need to analyze the MI of a single subchannel [7]. The MI $I(S; Y)$ is denoted as,

$$\begin{aligned} I(S; Y) &= H(S) - H(S|Y) = H(Y) - H(Y|S) \\ &= \sum_{m=1}^M \sum_Y p_m p(Y|S_m) \log \frac{p(Y|S_m)}{p(Y)}, \end{aligned} \quad (7)$$

where $H(Y|S)$ is the conditional entropy of Y given S , $\mathbf{p}(Y|S) \in \mathbb{R}^{N_Y \times M}$ represents the conditional probability of Y given S which satisfies $p(Y|S) = \exp(-|Y - S|^2 / \sigma^2) / \pi \sigma^2$, σ^2 is the communication channel noise power and N_Y denotes the number of Y samples generated by the Monte Carlo method.

III. PROBABILISTIC CONSTELLATION SHAPING

In this section, a novel PCS optimization problem is established to balance the communication and SAR imaging functions for UAV-ISAC systems. Then, the AO algorithm is proposed to handle the non-convex problem.

In order to enhance the SAR imaging performance, we limit the AIR and the input probability characteristics to minimize the EISL, that is,

$$\begin{aligned} \min_{p_m} \quad & \sum_{m=1}^M p_m |S_m|^4 - 1 \\ \text{s.t.} \quad & C_1 - C_3, C_4 : I(S; Y) \geq R, \end{aligned} \quad (8)$$

where R characterizes the minimum tolerable AIR and satisfies $0 < R \leq I(S; Y) < H_{\max}(S)$, $H_{\max}(S)$ denotes the symbol entropy in the equal probability case. Before proceeding with the problem solution, the constraint C_4 is first reformulated into the following computationally convenient form,

$$\begin{aligned} \bar{C}_4 : \quad & \max_{q(S_m|Y)} F(q(S_m|Y)) \geq R, \\ \bar{C}_5 : \quad & \sum_{m=1}^M q(S_m|Y) = 1, \forall Y, \end{aligned} \quad (9)$$

where $F(q(S_m|Y)) = \sum_{m=1}^M \sum_Y p_m p(Y|S_m) \log q(S_m|Y) / p_m$.

Then, the problem (8) can be rewritten as

$$\begin{aligned} \min_{p_m} \quad & \sum_{m=1}^M p_m |S_m|^4 - 1 \\ \text{s.t.} \quad & C_1, C_2, C_3, \bar{C}_4, \bar{C}_5. \end{aligned} \quad (10)$$

To solve this problem effectively, we introduce the AO framework to alternately update the variables $\mathbf{p}$ and $\mathbf{q}$.

A. Step 1 - Updating $\mathbf{p}^{(t)}$

Assuming that $\mathbf{q}^{(t-1)}$ is obtained in the $(t-1)$ -th iteration, the subproblem related to $\mathbf{p}$ in the t -th iteration can be simplified as,

$$\begin{aligned} \min_{p_m} \quad & \sum_{m=1}^M p_m |S_m|^4 - 1 \\ \text{s.t.} \quad & C_1 - C_3, \\ & \bar{C}_4 : \sum_{m=1}^M \sum_Y p_m p(Y|S_m) \log q^{(t-1)}(S_m|Y) \\ & - \sum_{m=1}^M p_m \log p_m \geq R. \end{aligned} \quad (11)$$

Note that the subproblem (11) is convex and can be solved via a variety of well-established algorithms, e.g., the interior-point method [11].

B. Step 2 - Updating $\mathbf{q}^{(t)}$

It is easy to observe that the subproblem related to $q(S_m|Y)$ includes constraints $\bar{C}_4$ and $\bar{C}_5$. Since $F(q^{(t)}(S_m|Y)) \geq F(q^{(t-1)}(S_m|Y)) \geq R$, the subproblem with respect to $q(S_m|Y)$ can be regarded as,

$$\begin{aligned} \max_{q(S_m|Y)} \quad & \sum_{m=1}^M \sum_Y p_m^{(t)} p(Y|S_m) \log \frac{q(S_m|Y)}{p_m^{(t)}} \\ \text{s.t.} \quad & \bar{C}_5 : \sum_{m=1}^M q(S_m|Y) = 1, \forall Y. \end{aligned} \quad (12)$$

According to [12], the solution to the above subproblem is

$$q^{(t)}(S_m|Y) = 0, \text{ if and only if } p(Y|S_m) = 0, \quad (13)$$

and

$$q^{(t)}(S_m|Y) = \frac{p_m^{(t)} p(Y|S_m)}{\sum_{m'=1}^M p_{m'}^{(t)} p(Y|S_{m'})}. \quad (14)$$

Finally, we repeat step 1 and step 2 until the maximum number of iterations $t_{\max}$ is exceeded, or the residual of the objective function satisfies

$$\left| \sum_{m=1}^M p_m^{(t)} |S_m|^4 - \sum_{m=1}^M p_m^{(t-1)} |S_m|^4 \right| \leq \xi, \quad (15)$$

where ξ denotes the convergence tolerance. For clarity, Algorithm 1 summarizes the main steps of the proposed AO algorithm. And the computational complexity in each iteration is $\mathcal{O}(M^{3.5}) + \mathcal{O}(N_Y M^2)$. It should be noted that the proposed PCS method is capable of generating a constellation probability lookup table to facilitate real-time applications.

Algorithm 1 AO Algorithm for Solving Problem (8)

Input: $S_m, R, N_Y, \xi, t_{\max}, \sigma^2$;

Output: $\mathbf{p}$;

- 1: Initialize $\mathbf{p}^{(0)}$ and $\mathbf{q}^{(0)}$;
 - 2: **for** $t = 1, \dots, t_{\max}$ **do**
 - 3: Step 1: Update $\mathbf{p}^{(t)}$ per Eq. (11);
 - 4: Step 2: Update $\mathbf{q}^{(t)}$ per Eq. (14);
 - 5: **if** Eq. (15) is satisfied **then**
 - 6: Break;
 - 7: **end if**
 - 8: **end for**
 - 9: Return $\mathbf{p} = \mathbf{p}^{(t)}$.
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IV. NUMERICAL RESULTS

In this section, numerical simulations are provided to illustrate the performance of the proposed PCS method when applied to UAV-ISAC systems. The parameters are set as follows: $N_Y = 10^4$, $\xi = 10^{-3}$ and $t_{\max} = 10^3$. Given the page limit, the 64-QAM constellation and the 64-PSK constellation are selected as representative cases for detailed analysis.

First, to evaluate the trade-off between communication and sensing functions, the variation trends of the objective function with respect to the AIR constraint R are compared under different noise power conditions $\sigma^2 = \{0.01, 0.02, 0.04, 0.06, 0.08, 0.1\}$. As shown in Fig. 3, the objective function exhibits a monotonic increase as the parameter R increases, which indicates that improving communication performance leads to a degradation in SAR imaging performance. Except for low-noise conditions, the proposed ISAC performance is traded off between uniform QAM and uniform PSK schemes. Furthermore, it is obvious that the value range of the objective function remains consistently confined to the interval $[0.036, 0.381]$. This is due to the fact that the proposed PCS algorithm fails to find a constant-modulus circle with $\kappa = 1$ for 64-QAM, where the minimum attainable kurtosis

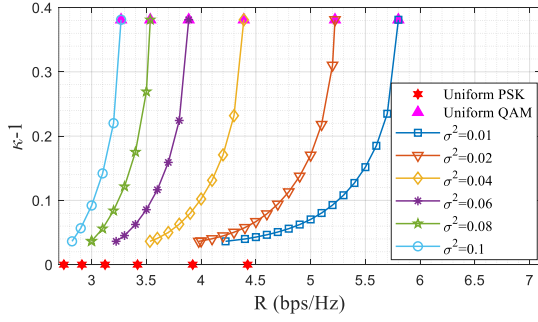


Fig. 3. ISAC performance trade-off: the objective function versus the parameter R under different noise power conditions.

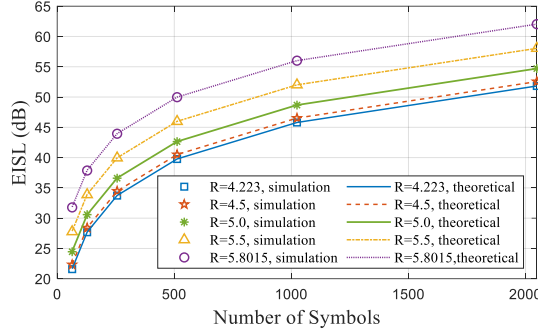


Fig. 4. The EISL versus the number of symbols.

is $\kappa = 1.036$, and the maximum kurtosis $\kappa = 1.381$ can be obtained in the uniform 64-QAM case.

Then, we extract several simulation results from the $\sigma^2 = 0.01$ curve in Fig. 3 to further analyze the P-ACF performance. Fig. 4 depicts variation trends of the theoretical and simulated EISL with the number of communication symbols, i.e., $K = \{64, 128, 256, 512, 1024, 2048\}$. Herein, the simulated EISL is derived by averaging the results from 1000 independent Monte Carlo simulation trials. Notably, $R = 4.223$ corresponds to the scenario with optimal imaging performance and $R = 5.8015$ represents the scenario with optimal communication performance, namely the uniform 64-QAM case. It is evident that all the simulated results closely match their corresponding theoretical results, and the EISL increases as the parameter R increases, resulting in degraded imaging quality.

Finally, we illustrate the SAR imaging performance of the proposed PCS method. The parameters are set as follows: $f_c = 9$ GHz, the bandwidth $B = 512$ MHz, $K = 512$, $H_U = 100$ m, $v_U = 10$ m/s, the synthetic aperture time is $T_a = 2$ s, the center of the target scene is located at (500 m, 0, 0) and the size of the target scene is 13.2×17.2 m. Figs. 5 (a), (c), (e) and (g) illustrate the constellation corresponding to the uniform 64-QAM, the optimized 64-QAM and the uniform 64-PSK scheme. To better illustrate the PCS effect, the non-zero probability values of the symbols are marked above their coordinates. And the corresponding 2-D SAR images with a signal to-noise ratio of 10 dB are shown in Figs. 5 (b), (d), (f) and (h). As evidenced by these figures, the intrinsic amplitude randomness of uniform 64-QAM scheme results in

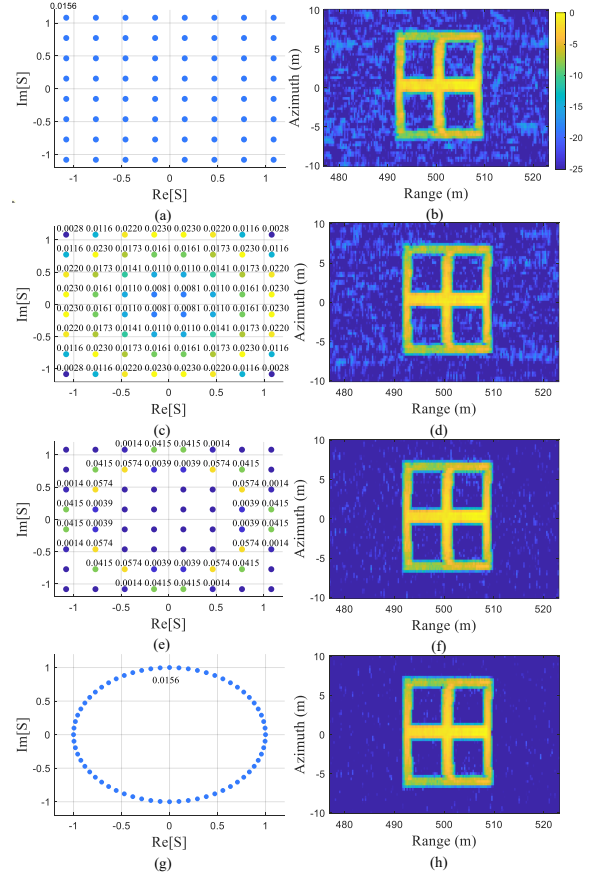


Fig. 5. SAR imaging performance: (a) the constellation corresponding to uniform 64-QAM $R = 4.3894$ bps/Hz, $\kappa = 1.3810$; (b) the SAR image corresponding to (a); (c) the constellation corresponding to $\sigma^2 = 0.04$, $R = 4.3$ bps/Hz, $\kappa = 1.2318$; (d) the SAR image corresponding to (c); (e) the constellation corresponding to $\sigma^2 = 0.04$, $R = 3.6$ bps/Hz, $\kappa = 1.0409$; (f) the SAR image corresponding to (e); (g) the constellation corresponding to uniform 64-PSK $R = 3.4189$ bps/Hz, $\kappa = 1.0$; (h) the SAR image corresponding to (g).

the poorest imaging performance, characterized by abundant noise artifacts in the imaging area outside of point targets. Meanwhile, the proposed PCS method enhances the imaging performance by reducing the randomness of the amplitude of QAM symbols, making the imaging results closer to that of PSK scheme.

V. CONCLUSION

In this paper, we have devised a PCS strategy under OFDM waveform for UAV-ISAC imaging system. In order to balance the effect of communication symbol amplitude randomness on SAR imaging performance, we proposed a multi-constrained PCS problem to minimize the EISL of the P-ACF under the constraints of the symbol probability and the AIR. To tackle the nonconvex problem, we have presented the AO algorithm to simplify the problem into convex subproblems, which can be solved iteratively. Finally, numerical simulations have verified the effectiveness of the proposed PCS approach in improving the SAR imaging performance. Future research will focus on the improvement of imaging algorithms.

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